

Solano County Health & Social Services Department

Mental Health Services
Public Health Services
Substance Abuse Services
Older & Disabled Adult Services



Patrick O. Duterte, Director

Eligibility Services
Employment Services
Children's Services
Administrative Services

Michael A. Frenn
EMS Agency Administrator

EMERGENCY MEDICAL SERVICES AGENCY
275 Beck Ave., MS 5-240
Fairfield, CA 94533
(707) 784-8155 FAX (707) 421-6682
www.solanocounty.com

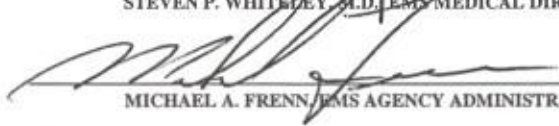
Steven P. Whiteley, M.D.
EMS Agency Medical Director

POLICY MEMORANDUM 6601

DATE: February 14, 2005

REVIEWED/APPROVED BY:


STEVEN P. WHITELEY, M.D., EMS MEDICAL DIRECTOR


MICHAEL A. FRENN, EMS AGENCY ADMINISTRATOR

SUBJECT: COMBITUBE

AUTHORITY: California Health & Safety Code §1797.220

I. INTENT

To offer an effective back up airway to endotracheal intubation. The preferred method of airway control is oral endotracheal intubation. In the event that the paramedic is unable to secure an airway by oral endotracheal intubation the Combitube is the back up airway device.

II. EDUCATION AND TRAINING:

Each agency shall provide Combitube training per the manufacturer's recommendations. All ALS providers will conduct annual training with the Combitube device.

III. INDICATIONS FOR USE:

- A. Cardiac and/or respiratory arrest and
- B. Not more than two (2) attempts at endotracheal intubation.

IV. CONTRAINDICATIONS FOR USE:

- A. Known esophageal disease.
- B. Ingestion of caustic substances.
- C. Positive gag reflex.

- D. Burns involving the airway.
- E. Persons under 4 feet in height.
- F. Presence of a tracheostomy or stoma.

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V. RELATIVE CONTRAINDICATIONS:

- A. Foreign body airway obstructions.
- B. Anatomical disruption of the oropharynx.
- C. Hypoglycemia
- D. Narcotic Overdose
- F. Severe facial/oral trauma

VI. EQUIPMENT:

- A. A complete Combitube kit of the appropriate size for the patient;
- B. 100 + cc syringe.
- C. 20 cc syringe;
- D. Water soluble lubricant;
- E. Stethoscope
- F. Portable suction;
- G. BVM;
- H. PPE (Personal Protective Equipment)

VII. INSERTION PROCEDURE:

- A. For patients 4 feet to 5 feet 6 inches, use the Combitube SA (short adult). For persons 5 feet and taller use the standard Combitube.
- B. Inflate and test each cuff, then deflate.
- C. Lubricate Combitube with water soluble lubricant;
- D. Hyperventilate patient;
- E. Place the head in a neutral position;

- F. Insert Combitube in the midline of the patient using a downward curved movement. Advanced until front teeth or alveolar ridges are between the black rings on the Combitube. Do not force the tube. A laryngoscope may be used to lift the tongue and jaw if desired.
- G. Inflate pharyngeal cuff with a 100ml of air (85ml for Combitube SA). The Combitube may move slightly as it seats in the pharynx – this is normal. Additional air may be inserted if needed to seal the airway during ventilation.
- H. Inflate the distal cuff with 15 ml of air (12ml for Combitube SA).
- I. Begin ventilations with a BVM device using the #1 port. Auscultate the patient's lung sounds. If auscultation reveals presence of breath sounds and the absences of gastric sounds continue ventilations via the #1 port. Emesis may issue from the #2 port. Reassess airway seals.
- J. If auscultation reveals absence of breath sounds and presence of gastric sounds immediately begin ventilations via port #2. Reassess breath sounds and airway seals.
- K. Confirm tube placement by using an approved confirmatory device, such as an end tidal CO₂ detector. Response to confirmation may be slower than ET intubation.
- L. Secure the tube and ventilate the patient with a BVM attached to 100% Oxygen.
- M. Reevaluate the Combitube's position each time after the patient moves to ensure the proper placement of the tube.

VIII. POTENTIAL COMPLICATIONS

- A. Subcutaneous emphysema;
- B. Perforated esophagus or trachea;
- C. Retropharyngeal perforation

IX. TROUBLESHOOTING AND ADDITIONAL INFORMATION

- A. Most unsuccessful placements relate to failure to place the tube in midline during placement.
- B. If placement is unsuccessful, remove tube, ventilate via BVM and repeat the sequence of steps.
- C. If no breath sounds are heard with ventilation through tube #1 or tube #2, it is possible that the tube has been placed too far into the pharynx. Deflate #1 pilot balloon and retract 2-3 cm, then reinflate cuff. Recheck lung sounds.
- D. Medications can only be given if the tube is placed into the trachea. Medications should not be administered with esophageal placement.
- E. Cuffs can be lacerated by broken teeth or dentures. Remove dentures before placing the tube.

X. CONTINUOUS QUALITY IMPROVEMENT

- A. Any adverse outcomes will be reported by Unusual Occurrence to the Medical Director of the Program, the Program CQI Coordinator and the EMS Agency within 24 hours. These calls will be investigated and the outcome of the investigation will be reported to the CQI Committee. The information to be reported is: Date of occurrence, the nature of the adverse outcome, what can be done to prevent the problem again.
- B. The ALS Provider will report at the CQI Committee Meeting the number of uses of the Combitube in the month and the circumstances surrounding the use.
- C. The ALS Provider will track individual paramedic statistics for use of the Combitube. This information will be available for the EMS Agency to review when requested.
- D. The ALS Provider will maintain records which show whether or not a paramedic has undergone training in the use of the Combitube. These records will be sent to the EMS Agency at the end of a calendar year, to be received no later than the last day of January of the new year.
- E. The ALS Provider will provide annual training for use of the Combitube to ensure paramedic competency. The training can be a subset of other airway training. It will include a skills demonstration of the Combitube by the paramedic.

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Richard Watson
Agency Administrator
(Interim)

POLICY MEMORANDUM 6602

DATE: 31 March 2010

REVIEWED/APPROVED BY: *December 17, 2009*

[Signature]
RICHARD C. LOTSCH, D.O., EMS MEDICAL DIRECTOR

[Signature]
RONALD W. CHAPMAN, M.D., EMS AGENCY ADMINISTRATOR (ACTING)

AUTHORITY: California Health & Safety Code §1797.204 & §1797.220

SUBJECT: 12-LEAD EKG

PURPOSE/POLICY:

In the prehospital evaluation of patients with chest pain, acquisition of a 12 lead EKG by field personnel provides information which is useful in answering a number of questions.

- A. Recognition of acute ST elevation myocardial infarction (STEMI) identifies a subgroup of patients who should benefit from the quickest possible transport to a hospital that can administer thrombolytic therapy or primary percutaneous coronary intervention. Pre-arrival notification of the receiving hospital is expected to significantly decrease delay in providing effective therapy.
- B. Chest pain patients without STEMI should be stable to be transported to the hospital where they typically receive their care, which is felt to be in the best interest of both patients and providers as long as it doesn't delay critical therapy.
- C. Transient 12 lead EKG findings of ischemia and arrhythmia which resolve in response to prehospital treatment can be very useful in arriving at an emergency department disposition.

- D. Patients with an acute STEMI MI will be transported to the nearest Solano County designated STEMI Receiving Center (SRC), see protocol C14 for SRCs.

I. TRAINING

Paramedics who use the 12-lead EKG shall have an orientation to the equipment and an additional 4 hours of training on the lead placement, general troubleshooting and interpretation of acute MI patterns. This class will be submitted for approval to the EMS Agency and will be updated biannually.

II. INDICATIONS

12 lead EKG is indicated for patients with chest pain/chest tightness or other symptoms which suggest Acute Coronary Syndrome (ACS); ischemic heart disease or acute myocardial infarction.

III. PROCEDURE

- A. Those EMT-Ps with the additional training and equipment will complete an EKG on patients as indicated above.
1. It is expected that non-transporting paramedics will immediately obtain or assist transporting paramedics to immediately obtain the 12 lead EKG
- B. The 12 lead EKG must be obtained in the field and will be used by the paramedic's to guide the patients' destination decision. If the 12 lead EKG reads *** Acute MI*** the paramedic will transport the patient to the closest SRC.
- C. In conjunction with current treatment, the EMT-P will place leads in the following positions:
1. Limb: RIGHT arm, RIGHT leg, LEFT arm, LEFT leg
 2. CHEST LEADS:
 - a). V₁ – RIGHT of Sternum at 4th intercostal space;
 - b). V₂ – LEFT of sternum at 4th intercostal space;
 - c). V₃ – ½ way between V2 & V4.
 - d). V₄ – LEFT chest, 5th intercostal space, mid-clavicular line.
 - e). V₅ – LEFT chest, 5th intercostal space on the anterior axillary line.
 - f). V₆ - LEFT chest, 5th intercostal space, mid-axillary line.
- D. Obtain EKG and report the computer interpretation of the 12-lead to the receiving hospital. If requested by the Base Hospital physician, a fax copy of the EKG will be transmitted when possible.
- E. EKG will be attached to the PCR and be identified with patient name, date and PCR#. Adhesive skin electrodes will be left in place on the patient to facilitate confirmation and to track evolving EKG changes

- F. The field 12-lead EKG is NOT meant to be diagnostic; instead to be used as an additional tool for the receiving hospital.

IV. CONTINUOUS QUALITY IMPROVEMENT

The CQI Coordinator for the ALS Departments and Medic Ambulance Service will, in conjunction with the CQI Committee, develop and enhance the current CQI policies for the use of 12-lead EKG.

- A. The ALS Departments CQI Coordinator's will evaluate the use of the 12 lead EKG on the patients.
- B. Each ALS provider will attach the original 12 Lead EKG to the patient's copy of the PCR and leave it at the receiving hospital, a copy of the 12 Lead EKG will be attached to original PCR and be kept as part of the medical records by the provider.
- C. Each agency using the 12-lead EKGs will complete annual update training of the paramedic using the machine. This information will be submitted to the EMS Agency by the end of the month of January of the new year.
- D. Each agency using the 12-lead EKGs will hold an annual initial training class for new employees and provide documentation to the EMS Agency when the training is completed and the paramedics who attended.
- E. The Base Coordinator will track cases where pre-hospital personnel used the 12 lead EKG for a patient with STEMI and report back to the CQI Committee if the patient received a thrombolytic agent and time "from door to drug" as a part of their monthly report.

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Gerald Huber, Director

Aaron E. Bair, MD, MS
EMS Agency Medical Director

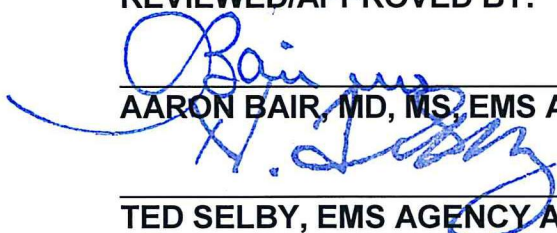
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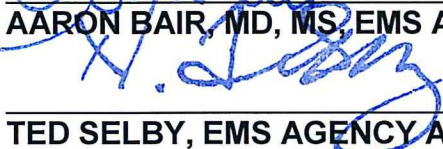
Ted Selby
EMS Agency Administrator

POLICY MEMORANDUM 6603

EFFECTIVE DATE: December 28, 2015

REVIEWED/APPROVED BY:


AARON BAIR, MD, MS, EMS AGENCY MEDICAL DIRECTOR


TED SELBY, EMS AGENCY ADMINISTRATOR

SUBJECT: INTRAOSSEOUS CANNULATION – ADULT OR PEDIATRIC

AUTHORITY: CALIFORNIA HEALTH & SAFETY CODE, DIVISION 2.5, 1797.200.

I. PURPOSE/POLICY:

To establish policy and procedure for the use of Intraosseous Infusion in adult and pediatric populations.

II. DEFINITIONS

- A. Intraosseous Cannulation (IO): The use of a manual or mechanical device to insert an appropriate needle through the bone into marrow cavity for medication administration or fluid resuscitation.
- B. EZ-IO: Mechanical device recommended for use in adult and pediatric populations for IO Cannulation.
- C. Bone Injection Gun or B.I.G.: Alternative mechanical device for use in adult and pediatric populations for IO Cannulation.
- D. Manual Intraosseous Infusion Needle: Only for use in pediatrics less than six years of age.

III. CRITERIA FOR USE

- A. Pediatrics:
 - 1. Patient critically ill or injured when IV fluids and/or medications must be administered to prevent death;
 - 2. Unresponsive;
 - 3. Venous access unavailable within 90 seconds.
- B. Adults:
 - 1. Patient critically ill or injured when IV fluids and/or medications should be immediately administered;
 - 2. May also be used after two unsuccessful IV attempts.

IV. CONTRAINDICATIONS

- A. Fracture proximal to placement site;
- B. Previous orthopedic procedures near insertion site;
- C. Inability to locate landmarks for insertion;
- D. Pre-existing medical condition of extremity;
- E. Infection at insertion site;
- F. Excessive pre-tibial or pre-humeral soft tissue or swelling;
- G. Previous IO within 24 hours;

V. EQUIPMENT

- A. Intraosseous infusion needle (Pediatric);
- B. EZ-IO DRIVER (Drill Device);
- C. EZ-IO Needle: 25 mm (adult and pediatric patients >3Kg) and 45 mm (adult humeral or >40 Kg with excessive tissue depth).
- D. Bone Injection Gun or B.I.G. (Adult and Pediatric sizes available);
- E. Appropriate county-recommended antiseptic cleanser;
- F. Sterile Gauze pads;
- G. 10 – 12 mL syringe filled with normal saline;
- H. Supplies to secure infusion.

VI. PROCEDURE FOR INTRAOSSEOUS CANNULATION

- A. Wear appropriate personal protective equipment;
- B. Note: Insertion sites are the same for the EZ-IO and Bone Injection Gun; See attachment A for Tibial placement and attachment B for Humeral placement.
- C. Using aseptic technique, cleanse the insertion site with appropriate county-recommended antiseptic cleanser;
- D. Stabilize the leg. For humeral insertion secure the arm in place across the abdomen.

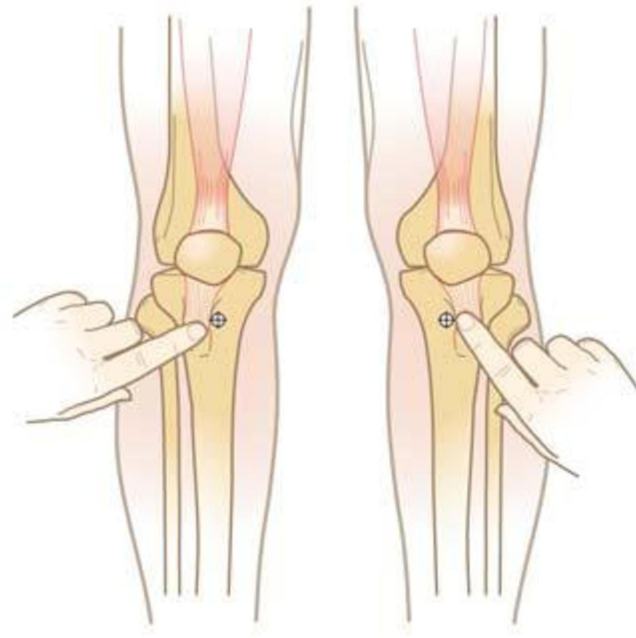
- E. Locate the insertion site and Insert intraosseous needle. The proximal humerus is the preferred insertion site for adult patients.
 - 1. Humeral-
 - a. Place the patient's hand over the abdomen (elbow adducted and humerus internally rotated)
 - b. Place your palm on the patient's shoulder anteriorly; the "ball" under your palm is the general target area
 - c. You should be able to feel this ball, even on obese patients, by pushing deeply
 - d. Place the ulnar aspect of your hand vertically over the axilla and the ulnar aspect of your other hand along the midline of the upper arm laterally
 - e. Place your thumbs together over the arm; this identifies the vertical line of insertion on the proximal humerus
 - f. Palpate deeply up the humerus to the surgical neck
 - g. This may feel like a golf ball on a tee – the spot where the "ball" meets the "tee" is the surgical neck
 - h. The insertion site is 1 to 2 cm above the surgical neck, on the most prominent aspect of the greater tubercle
 - 2. Tibia-
 - a. Aim the needle set at a 90-degree angle to the bone
 - b. Push the needle set tip through the skin until the tip rests against the bone
 - c. The 5 mm mark must be visible above the skin for confirmation of adequate needle set length
 - d. Gently drill, advancing the needle set approximately 1-2 cm after entry into the medullary space or until the needle set hub is close to the skin
- F. Flush or bolus the IO catheter rapidly with 10 mL of normal saline using a 10 mL syringe; for pediatrics use 2-5 mL normal saline.
- G. Confirm proper placement in marrow cavity as evidenced by:
 - 1. A decrease in resistance after needle passes through the bone cortex;
 - 2. The needle standing upright without support;
 - 3. The ability to aspirate bone marrow into syringe connected to the needle;
 - 4. Ability to push IV fluids through the IO needle without evidence of subcutaneous infiltration;
- H. Dress the site and secure the IO device in place;
- I. Lidocaine 2% administration for pain relief in conscious patients;
 - 1. Prepare Lidocaine 2% -40 mg for adults and 0.5 mg/kg for pediatrics(max dose of 40mg)
 - 2. Prime extension set with Lidocaine

3. Note that the priming volume of the EZ-Connect® Extension Set is approximately 1.0 mL
 4. Slowly infuse lidocaine 40 mg IO over 120 seconds
 5. Allow Lidocaine to dwell in IO space 60 seconds
- J. Since there is relatively high pressure in the bone marrow cavity, fluids and medications **must** be administered under pressure when given using an intraosseous access device. This requires the use of either a pressure bag on the IV solution or the use of a syringe to inject fluids or medications under pressure.
- K. Continuous monitoring of soft tissues surrounding insertion site is required for early recognition of signs of infiltration.

VII. CONTINUOUS QUALITY IMPROVEMENT (CQI)

- A. Each agency's CQI Coordinator will evaluate the use of the IO on all patients (adult and pediatric) and report results at County CQI committee meetings;
- B. Each Agency using the IO device for adults will provide initial training for all paramedics to ensure competency in the procedure;
- C. To ensure skill competency each Advanced Life Support (ALS) Provider will provide annual "skills" practical update training for adult and pediatric IO;
- D. Annual training records for adult and pediatric IO will be submitted to the EMS Agency by the end of January each year for training accomplished the preceding calendar year.

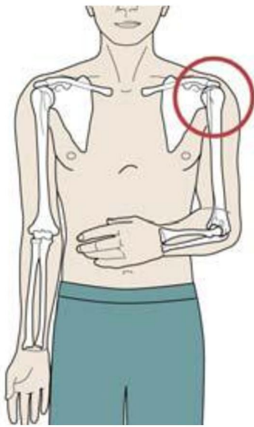
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Proximal Tibia

The proximal tibia insertion site is approximately 2 centimeters below the patella and approximately 2 centimeters medial to the tibial tuberosity (depending on patient anatomy).

Insertion site identification summary



Orient Arm



Preferred Approach



Alternate Approach

The patient should be in a supine position

1. Expose shoulder, adduct humerus (place the patient's arm against the patient's body) resting the elbow on the stretcher or ground, and have the forearm resting on the abdomen.
2. Palpate and identify the mid-shaft humerus and continue palpating toward the proximal aspect or humeral head. A small protrusion exists near the shoulder; this is the base of the greater tubercle insertion site. With the opposite hand consider "pinching" the anterior and inferior aspects of the humeral head while confirming the identification of the greater tubercle. This will ensure that the midline of the humerus itself has been identified.

Alternatively:

1. Identify two land marks on the lateral shoulder consisting of the acromion and the coracoid process. This can be accomplished by placing one hand on the lateral superior aspect of the patient's shoulder and palpating for the protrusions. Identifying the coracoid process and the acromion can also be accomplished by "walking" the index and middle fingers along the clavicle to the shoulder's lateral end.
2. Identify the greater tubercle insertion site approximately two finger widths inferior to the coracoid process and the acromion. To envision the location of this site, create a "T" - the upper portion of the letter connecting the coracoid process and the acromion while the "leg" reaches inferiorly and slightly anteriorly - approximately two finger widths along the midline between the two structures.

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**SOLANO
COUNTY**

TED SELBY
EMS Agency Administrator

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POLICY MEMORANDUM 6605

Implementation Date: November 30, 2009

Revised Date: January 20, 2020

Review Date: January 20, 2022

REVIEWED/APPROVED BY:

 MD, MAS

BRYN MUMMA, MD, MAS, EMS AGENCY MEDICAL DIRECTOR



TED SELBY, EMS AGENCY ADMINISTRATOR

SUBJECT: CONTINUOUS POSITIVE AIRWAY PRESSURE (CPAP)

AUTHORITY:

California Health and Safety Code Sections 1797.220 and 1797.221
California Code of Regulations Sections 100063 and 100146

PURPOSE:

To improve ventilation and oxygenation and avoid intubation for patients suffering from Congestive Heart Failure (CHF), Chronic Obstructive Pulmonary Disease (COPD), asthma, pneumonia, generalized Shortness of Breath (SOB), or near drowning and are at risk for acute pulmonary edema or bronchospasm.

I. INDICATIONS

- A. For patients ≥ 12 years old in severe respiratory distress from the following conditions:
1. Acute pulmonary edema and CHF;
 2. Near drowning;
 3. Asthma;
 4. Bronchospasm;
 5. COPD;
 6. Pneumonia;
 7. Non-traumatic SOB;

II. CONTRAINDICATIONS

- A. CPAP is contraindicated with any of the following:
 - 1. Patient ≤ 11 years old;
 - 2. Respiratory or cardiac arrest;
 - 3. Agonal respirations;
 - 4. Severely depressed level of consciousness with a Glasgow Coma Scale of ≤ 8 ;
 - 5. Systolic blood pressure < 90 mmHg;
 - 6. Pneumothorax;
 - 7. Inability to maintain airway patency;
 - 8. Major trauma, especially with head and/or chest trauma;
 - 9. Facial abnormalities;
 - 10. Vomiting;

III. COMPLICATIONS FROM CPAP

- A. Complications from CPAP are any of the following:
 - 1. Hypotension;
 - 2. Pneumothorax;
 - 3. Corneal drying;
 - 4. Patient agitation or inability to tolerate CPAP;

IV. CPAP PROCEDURE

- A. Evaluate the following prior to applying CPAP:
 - 1. Heart rate;
 - 2. Respiratory Rate;
 - 3. Blood Pressure;
 - 4. Pulse Oximetry;
 - 5. Mental status;
 - 6. Lung sounds;
 - 7. Work of breathing;
- B. Explain the procedure to the patient.
- C. Set up the CPAP per manufacturer's recommendations and set to 10 cmH₂O pressure.
- D. Apply CPAP and encourage patient to breathe normally. Check mask seal to ensure that no air is leaking.
- E. Reassess patient every five minutes using the criteria in Section IV(A).

- F. Notify the receiving hospital as soon as possible that CPAP has been utilized.
- G. Do not remove CPAP until the receiving hospital is prepared to switch the patient to the hospital CPAP or unless ordered by the Base Hospital Physician.

V. GOALS OF CPAP

- A. To improve the patient's respiratory status by reducing the work of breathing or improving suspected hypercapnia/hypoxemia to avoid more invasive airway management.
 - 1. If the patient fails to improve, consider bag-valve mask ventilation and endotracheal intubation for adult patients and Basic Life Support (BLS) airways for pediatric patients.
-